

PfMSP3.1 SPAM Structural Characterisation

Detailed repository report | Historical 2020 Master's research record and 2026 computational reassessment

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Executive summary

This report documents a public computational research repository preserving the structural and functional characterisation of the SPAM domain of Plasmodium falciparum merozoite surface protein 3.1 (PfMSP3.1). The repository separates the original 2020 Master's research record from reconstructed historical results, exploratory work and a sequence-matched 2026 structural reassessment. Its purpose is preservation, transparency and reproducibility; it is not a peer-reviewed manuscript and does not present the historical model as a newly validated structure.

1 Project purpose and background

Plasmodium falciparum is the parasite responsible for the most severe form of human malaria. The 2020 Master's project investigated the PfMSP3.1 SPAM region using structural bioinformatics to examine coiled-coil organisation, model stability and candidate interactions with the human complement regulator C1-inhibitor (C1-INH). The 2026 work preserved surviving files, reconstructed documented thesis outputs where possible, and reassessed the historical structural interpretation using contemporary AlphaFold 3 predictions matched to the same 139-amino-acid sequence and the underlying experimental NMR restraints.

The repository deliberately distinguishes historical material from later analyses. Reconstructed tables preserve values reported in the thesis and should not be interpreted as newly rerun analyses. Exploratory analyses based on the 121-residue AlphaFoldDB entry A0A8G1DNL0 are retained for provenance but are not treated as direct validation because that sequence does not match the historical 139-residue construct.

2 Methods and software

Analysis area	Methods and principal software
Sequence and coiled-coil assessment	Sequence analysis, conservation assessment, helical-wheel/coiled-coil interpretation and CCBUILDER-assisted model construction.
Comparative modelling	MODELLER generation and scoring of candidate structures; structural assessment included objective-function/DOPE reporting and Ramachandran analysis.
Molecular dynamics	Historical triplicate 500-ns simulations using GROMACS and CHARMM36; analyses included RMSD, structural comparison and salt-bridge behaviour. A separate exploratory 250-ns 2026 simulation used GROMACS 2026.0 for preparation, GROMACS 2020.7 for HPC production and CHARMM36 July 2022.
Protein-protein docking	Docking against C1-INH structures, ZDOCK candidate generation, ZRANK rescoring, buried surface area analysis and MD assessment of selected complexes.
AF3/NMR reassessment	Five sequence-matched AlphaFold 3 predictions; experimental restraint audit; OpenMM hydrogenation of working copies; legacy atom-name mapping; evaluation of direct, pseudoatom and stereospecific restraints; consensus-violation assessment and C-alpha structural alignment.
Visualisation and inspection	UCSF Chimera, PyMOL and VMD were used across modelling, trajectory inspection and structural visualisation workflows.

3 Repository structure

Path	Contents
01_original_2020_work	Preserved sequence-analysis and other original historical material.
02_structural_modelling	CCBuilder and MODELLER inputs, scripts, candidate models and supporting documentation.
03_exploratory_AlphaFold_comparison	Exploratory assessment of the non-matching 121-residue AlphaFoldDB entry.
04_AF3_vs_experimental_NMR	Sequence-matched 2026 reassessment: provenance, NMR restraint data, scripts, derived results and final interpretation.
05_MD_simulations / 06_MD_analysis	Historical and 2026 simulation setup, retained inputs, logs, structures and derived analyses.
07_protein_docking / 08_complex_MD	Historical docking, rescoring, surface-area analysis and selected-complex MD material.
09_figures / 10_tables	Historical thesis figures, reassessment figures and reconstructed numerical thesis tables.
docs / reproducibility	Inventories, provenance notes and instructions linking historical and modern work.

4 Main analyses and results

The historical branch generated and assessed comparative structural models, examined three independent MD replicates, analysed salt-bridge behaviour, ranked docked PfMSP3.1-C1-INH complexes using ZRANK, compared buried surface area before and after MD, and retained the reported numerical outputs as reconstructed thesis tables.

The sequence-matched 2026 reassessment addressed whether AlphaFold 3 was incompatible with the original NMR evidence and whether the historical model resolved an experimental inconsistency. All five AF3 models satisfied all 14 evaluated i to $i+4$ distance restraints. Structural alignment between AF3 and the historical model showed close local agreement: C-alpha RMSDs were approximately 0.774-0.887

Angstrom within the experimentally relevant SPAM-H1 region and 0.292-0.406 Angstrom within residues 16-41, the principal experimental helical region.

Some apparent violations occurred in the broader distance-restraint audit of individual static structures. These were not treated automatically as contradictions because NMR-derived distances reflect conformational ensembles, particularly in flexible regions. The reassessment did not identify compelling evidence that the historical model provides a superior explanation of the experimental NMR observations. Following expert review, the historical-model branch was not pursued further as a new publication.

5 Limitations

- The repository is a computational research record, not a peer-reviewed publication or independent experimental validation study.
- Not every command, intermediate file or software environment from the 2020 work could be reconstructed; missing information was not inferred.
- Reconstructed thesis tables preserve reported historical values but do not represent complete reruns of the original workflow.
- Large raw trajectories and some generated outputs are stored outside GitHub because of file-size constraints; a complete public raw-data archive and checksum manifest are not yet available.
- The exploratory A0A8G1DNL0 analyses concern a non-matching sequence and cannot validate or reject the historical 139-residue model.
- Static-structure comparisons cannot fully represent conformational ensembles underlying NMR measurements.
- The externally supplied AF3 package and exact AF3 software build are not redistributed or independently recorded in the public repository.
- Docking predictions and candidate interfaces do not demonstrate a biological interaction without experimental confirmation.

6 Reproducibility instructions

Clone the repository and begin with the root README and the reproducibility guide. Treat each branch according to its provenance label before attempting reproduction or interpretation.

```
git clone https://github.com/dalalalhabad/PfMSP3.1-SPAM-Structural-Characterisation.git
cd PfMSP3.1-SPAM-Structural-Characterisation
```

- Read reproducibility/README.md to distinguish original 2020 files, reconstructed outputs, exploratory 2026 analyses and the sequence-matched reassessment.
- For the AF3/NMR reassessment, inspect inputs under 04_AF3_vs_experimental_NMR/input, scripts under scripts and derived outputs under results. Run numbered scripts in order where dependencies apply.
- Preserve original inputs. Create separate working copies for hydrogenation, renaming, alignment or re-analysis.
- Record software versions, operating system, commands and checksums for any new run. Do not retrospectively assign versions to historical steps when they cannot be verified.
- Interpret the 2020 reconstructed tables together with their local README files; do not label them as newly regenerated.

- Retrieve externally retained trajectories or AF3 source packages only through authorised project access and respect third-party licensing conditions.

7 Access and persistent identifier

GitHub: github.com/dalalalhabad/PfMSP3.1-SPAM-Structural-Characterisation

Zenodo DOI: [10.5281/zenodo.22763191](https://doi.org/10.5281/zenodo.22763191)

8 Recommended citation

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Users should also cite the relevant original experimental structures, datasets and software when relying on third-party material or reproducing a specific analysis.

9 Licence and reuse

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Prepared from the repository documentation and release record current on 15 September 2026.